

Math 871 Topology I Worksheet 5 solutions

Product topology

1. The universal property of the product topology: given a family of topological spaces $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in I}$, endow the cartesian product

$$\prod_{\alpha \in I} X_\alpha$$

with the product topology. Given a topological space S and a family of functions $f_\alpha: Y \rightarrow X_\alpha$, consider the function

$$\begin{aligned} f: S &\longrightarrow \prod_{\alpha \in I} X_\alpha \\ y &\longmapsto (f_\alpha(y))_{\alpha \in I}. \end{aligned}$$

Show that f is continuous if and only if f_α is continuous for all $\alpha \in I$.

Solution. ($\Rightarrow$) Let

$$\pi_\beta: \prod_{\alpha \in I} X_\alpha \rightarrow X_\beta$$

denote the projection onto X_β . By construction of the product topology, each π_α is continuous. If f is continuous, then $f_\alpha = \pi_\alpha \circ f$ is a composition of continuous functions and thus continuous for all $\alpha \in I$.

($\Leftarrow$) Suppose that f_α is continuous for all α . It suffices to check $f^{-1}(U)$ is open for every open set in a basis for the product topology, so we only need to consider open sets of the form

$$U = \prod_{\alpha \in I} U_\alpha \quad \text{with } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha \in I.$$

Suppose that

$$\{\alpha \in I \mid U_\alpha \neq X_\alpha\} = \{\alpha_1, \dots, \alpha_n\} =: J.$$

Now note that

$$s \in f^{-1}(U_\alpha) \iff f(s) = (f_\alpha(s))_{\alpha \in I} \in U \iff f_\alpha(s) \in U_\alpha \text{ for all } \alpha \in I.$$

When $\alpha \notin J$, $f_\alpha(s) \in U_\alpha = X_\alpha$ is automatically satisfied. Thus

$$s \in f^{-1}(U_\alpha) \iff f_{\alpha_i}(s) \in U_{\alpha_i} \text{ for } i = 1, \dots, n.$$

Thus

$$f^{-1}(U_\alpha) = \bigcap_{i=1}^n f_{\alpha_i}^{-1}(U_{\alpha_i})$$

is open since it is an intersection of finitely many open sets, as each f_α is continuous and each U_α is open in X_α .

2. Consider the diagonal function

$$\begin{aligned}\Delta: \mathbb{R} &\longrightarrow \mathbb{R}^\omega \\ t &\longmapsto (t) = (t, t, t, \dots).\end{aligned}$$

Show that Δ is continuous when $\mathbb{R}^\omega$ is given the product topology, but not under the box topology.

Solution. We can think of Δ as $\Delta = (\text{id})_{n \geq 1}$, where we take the identity function $\text{id}: \mathbb{R} \rightarrow \mathbb{R}$ on each copy of $\mathbb{R}$. If we endow each copy of $\mathbb{R}$ within sight with the standard topology, then Δ is continuous under the product topology on $\mathbb{R}^\omega$ by the universal property of the product topology from problem 1. However, we claim that Δ is not continuous if instead we endow $\mathbb{R}^\omega$ with the box topology. Indeed, the set

$$U = \prod_{i=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right)$$

is open in the box topology, but

$$\Delta^{-1}(U) = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$$

is not open in $\mathbb{R}$ under the standard topology.

3. Let $\{X_\alpha\}_{\alpha \in I}$ be any collection of topological spaces and endow

$$\prod_{\alpha \in I} X_\alpha$$

with the product topology. Let A_α be a subset of X_α for each $\alpha \in I$.

(a) Show that if every X_α is Hausdorff, then the product $\prod_{\alpha \in I} X_\alpha$ is Hausdorff.

Solution. Consider any two distinct elements $x = (x_\alpha)_{\alpha \in I}$ and $y = (y_\alpha)_{\alpha \in I}$ in the product. Then there exists $\beta \in I$ such that $x_\beta \neq y_\beta$.

Since X_β is Hausdorff, we can find neighborhoods A of x_β and B of y_β such that $A \cap B = \emptyset$. Set

$$U = \prod_{\alpha \in I} U_\alpha \text{ with } U_\alpha = \begin{cases} A & \text{if } \alpha = \beta \\ X_\alpha & \text{if } \alpha \neq \beta \end{cases}$$

and

$$V = \prod_{\alpha \in I} V_\alpha \text{ with } V_\alpha = \begin{cases} B & \text{if } \alpha = \beta \\ X_\alpha & \text{if } \alpha \neq \beta. \end{cases}$$

By definition of the product topology, U and V are open sets in the product. Moreover, if $(z_\alpha) \in U \cap V$, then in particular $z_\beta \in A \cap B = \emptyset$, so $U \cap V = \emptyset$.

- (b) Show that if all the A_α are closed, then their product is closed in the product topology.

Solution. Let

$$x = (x_\alpha)_{\alpha \in I} \in \prod_{\alpha \in I} X_\alpha \quad \text{with } x \notin \prod_{\alpha \in I} A_\alpha =: A.$$

Then $x_\beta \notin A_\beta$ for some $\beta \in I$. Since A_β is closed, there exists an open set V in X_β such that $x_\beta \in V$ and $V \cap A_\beta = \emptyset$. Thus

$$U = \prod_{\alpha \in I} U_\alpha \quad \text{with } U_\alpha = \begin{cases} V & \text{if } \alpha = \beta \\ X_\alpha & \text{if } \alpha \neq \beta \end{cases}$$

is an open neighborhood of x and $U \cap A = \emptyset$.

- (c) Show that

$$\prod_{\alpha \in I} \overline{A_\alpha} = \overline{\prod_{\alpha \in I} A_\alpha}.$$

Solution. ($\supseteq$) Since $A_\alpha \subseteq \overline{A_\alpha}$ for all α ,

$$C := \prod_{\alpha \in I} A_\alpha \subseteq \prod_{\alpha \in I} \overline{A_\alpha}.$$

Since the product of closed sets is closed,

$$\prod_{\alpha \in I} \overline{A_\alpha}$$

is a closed set. Since it contains C , it must also contain its closure. We conclude that

$$\overline{\prod_{\alpha \in I} A_\alpha} \subseteq \prod_{\alpha \in I} \overline{A_\alpha}.$$

($\subseteq$) Set

$$C := \prod_{\alpha \in I} A_\alpha.$$

Let

$$x = (x_\alpha)_{\alpha \in I} \in \prod_{\alpha \in I} \overline{A_\alpha}$$

so that $x_\alpha \in \overline{A_\alpha}$ for all $\alpha \in I$. Let U be any open neighborhood of x in our favorite basis, so that U is a product

$$U = \prod_{\alpha \in I} U_\alpha$$

of open sets U_α in X_α where $U_\alpha = X_\alpha$ for all but finitely many $\alpha \in I$. Then since $x_\alpha \in \overline{A_\alpha}$ for all $\alpha \in I$, $U_\alpha \cap A_\alpha \neq \emptyset$. Taking $a_\alpha \in U_\alpha \cap A_\alpha$ for each α ,

$$a = (a_\alpha)_{\alpha \in I} \in U \cap C.$$

We conclude that $a \in \overline{C}$.

(d) Consider

$$A = \prod_{\alpha \in I} A_{\alpha}.$$

Show that the product topology on A induced by giving every A_{α} the subspace topology agrees with the subspace topology of A as a subspace of $\prod_{\alpha} X_{\alpha}$ with the product topology.

Solution. We get a basis for the subspace topology on A by intersecting the elements of a basis for the product topology on the entire space with A . Thus

$$\mathcal{B}_{\text{sub}} = \left\{ \prod_{\alpha \in I} A_{\alpha} \cap \prod_{\alpha \in I} U_{\alpha} \mid \begin{array}{l} U_{\alpha} \text{ is open in } X_{\alpha} \\ U_{\alpha} = X_{\alpha} \text{ for all but finitely many } \alpha \in I \end{array} \right\}$$

is a basis for the subspace topology. Similarly, we get a basis for the subspace topology on each A_{α} by taking

$$\{U_{\alpha} \cap A_{\alpha} \mid U_{\alpha} \text{ open in } X_{\alpha}\}.$$

Note that $A_{\alpha} = X_{\alpha} \cap A_{\alpha}$ is also of this form. We then get a basis for the product topology by taking

$$\mathcal{B}_{\text{prod}} = \left\{ \prod_{\alpha \in I} (U_{\alpha} \cap A_{\alpha}) \mid \begin{array}{l} U_{\alpha} \text{ is open in } X_{\alpha} \\ U_{\alpha} = X_{\alpha} \text{ for all but finitely many } \alpha \in I \end{array} \right\}.$$

Continuous functions

6. Consider $\mathbb{R}$ under the standard topology. Give an example of a function $f: \mathbb{R} \rightarrow \mathbb{R}$ and a countable collection of closed subsets $\{A_{\alpha}\}_{\alpha \in I}$ such that the following all hold:

- The restrictions $f|_{A_{\alpha}}$ are continuous for all $\alpha \in I$.
- The closed sets A_{α} cover $\mathbb{R}$, meaning $\bigcup_{\alpha \in I} A_{\alpha} = \mathbb{R}$.
- The function f is not continuous.

Solution. Let f be given by

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0. \end{cases}$$

The set $U = (-\frac{1}{2}, \frac{1}{2})$ is open but $f^{-1}(U) = \{0\}$ is not, so f is not continuous.

On the other hand, taking $A_0 = \{0\}$ and

$$A_n = \left[-\infty, \frac{1}{n}\right] \cup \left[\frac{1}{n}, \infty\right]$$

for all integers $n \geq 1$, we get a collection of closed sets $\{A_n\}_{n \geq 0}$ with

$$\bigcup_{\alpha \in I} A_\alpha = \mathbb{R},$$

and such that $f|_{A_n}$ is a constant function for all $n \geq 0$, which is thus continuous.

7. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be topologies on X . Show that the identity function

$$\text{id}: (X, \mathcal{T}_1) \rightarrow (X, \mathcal{T}_2)$$

is continuous if and only if $\mathcal{T}_1$ is finer than $\mathcal{T}_2$.

Solution. For any subset $U \subseteq X$,

$$\text{id}^{-1}(U) = U.$$

Thus

$$\text{id is continuous} \iff \text{every } U \in \mathcal{T}_2 \text{ satisfies } U = \text{id}^{-1}(U) \in \mathcal{T}_1 \iff \mathcal{T}_2 \subseteq \mathcal{T}_1.$$

8. Consider $\mathbb{R}$ under the standard topology and fix real numbers $a < b$. Show that the subspace (a, b) is homeomorphic to $(0, 1)$ and the subspace $[a, b]$ is homeomorphic to $[0, 1]$.

Solution. Consider the function $f: (a, b) \rightarrow (0, 1)$ and $g: (0, 1) \rightarrow (a, b)$ given by

$$f(x) = a + (b - a)x \quad \text{and} \quad g(x) = \frac{x - a}{b - a}.$$

Note that

$$fg(x) = a + (b - a) \left(\frac{x - a}{b - a} \right) = x$$

and

$$gf(x) = \frac{a + (b - a)x - a}{b - a} = x$$

so f is bijective with inverse g .

To check continuity, we use a homework problem to prove that any linear function $f: \mathbb{R} \rightarrow \mathbb{R}$, that is, with

$$f(x) = ax + b$$

for some $a, b \in \mathbb{R}$ is actually continuous. On Problem Set 7, we proved that the sum and product of continuous functions is continuous. Moreover, we proved in class that constant functions are continuous, and by problem 8 the identity function is continuous whenever the domain and codomain are given the same topology. Now we see that $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = ax + b$$

is the sum of a constant function (continuous) with the product of a constant function by the identity functions, and thus it is continuous. We conclude that our original functions f and g are both continuous, as they are linear.